GENERAL SPECIFICATION G3.25

Protective Linings for Concrete (Tank Internal)

CONTENTS

G3.25 PROTECTIVE LININGS FOR CONCRETE (TANK INTERNAL)

G3.25.1 General

(a) References

(i) The following is a list of standards which may be referenced in this Section:

1) American Society for Testing and Materials (ASTM)

- C794, Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants.
- D2240, Test Method for Rubber Property - Durometer Hardness.

2) NACE International: RP0188-99, Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates

3) Steel Structures Painting Council (SSPC)

- SP 1, Surface Preparation Specification No. 1, Solvent Cleaning.
- SP 5, White Metal Blast Cleaning.
- SP 13, Surface Preparation of Concrete.

(b) Description of Work

(i) Plastic sheet lining systems shall be provided for corrosion protection of interior surfaces of concrete wastewater structures where shown in the drawings.

(c) Qualifications for Lining System

(i) The lining system shall be from the standard product range of an approved specialist lining system manufacturer, who shall be employed to install the lining system. The Contractor shall demonstrate that the lining system and lining system manufacturer have a minimum of five years successful experience in the installation/application of the proposed system. Only tradesmen experienced with the installation/application of the materials proposed lining system shall be employed for installation and commissioning.

(ii) The Contractor shall provide evidence that the lining system has a successful track record on similar projects for similar applications. In particular, the lining material must be entirely suitable for long term deployment in the areas to be protected. Among other things, depending on the specific application lining systems will be exposed to used water, sludge, gases (including hydrogen sulphide) and chemical solutions and other corrosive and erosive mater normally expected in the treatment of used water. The lining system examples cited shall have been in service for a minimum of five years without evidence of failure.

(iii) Samples: As a minimum, submit the following items:

1) Sample Panel

- Submit samples of complete lining system on 250mm x 250mm panel of substrate. Include standard joint, corner, and termination in the samples;

- Panel to be one of the reference sources for inspection; and
- Submit additional samples as required to show details of lining system.

(iv) Quality Control Submittals: As a minimum, submit the following items:

1) Manufacturer's Product History: Provide a list of not less than 5 projects where lining system has been installed. List shall include project name, date of installation, employer, contact person and phone number;

2) Certifications: Provide written statements of training and experience for individuals who will perform the installation; and

3) Single Manufacturer: All products used in lining system shall be products of the same manufacturer. If any product by a different manufacturer is proposed, submit all manufacturers' written confirmation that materials are compatible.

(d) Manufacturer's Services

(i) In accordance with the general specifications, manufacturer's representative shall be present at site full time during installation, testing and commissioning.

(e) Quality Assurance

(i) Qualifications of Applicators: Minimum 5 years' experience in application of specified products.

(ii) Regulatory Requirements

1) Conform to all requirements limiting emissions of dust and volatile organic compounds; and

2) Conform to all requirements for worker safety.

(iii) Mock-Up

1) Before proceeding with work, finish one complete space showing finish texture, materials, quality of work, and special details;

2) Conduct adhesion testing and spark testing of mock-up area to demonstrate testing equipment and compliance with the specified requirements; and

3) After approval, sample spaces or items shall serve as a standard for the installation of the lining system.

(f) Environmental Requirements

(i) Do not apply primers or adhesives when work will be done under ambient conditions, and such conditions are outside of manufacturer's recommended maximum or minimum allowable.

(ii) Provide environmental controls including heating, cooling or dehumidification as required when work will proceed and ambient conditions are outside manufacturer's recommended range.

(g) Lining System Extended Warranty

(i) Provide warranty with the Board named as the beneficiary for material as specified in the general specifications.

(ii) Guarantee stating that the lining system will be able to withstand the service in use, is corrosion resistant and free from defects for a period of not less than ten years from date of substantial completion of the Works shall be provided. In the event any tear or deterioration occur exposing any adhesive or concrete within the period stipulated, the Contractor shall, at the convenience of the Board, effect all repairs and replacements necessary to remedy defects all to the complete satisfaction of the S.O. at no additional cost to the Board.

G3.25.2 **Products**

(a) Materials

(i) The materials to be used shall be suitable for use in the locations specified.

(b) Sheet Lining System

(i) System consists of primer, adhesive mastic, activator, and plastic sheets and shapes (where required).

(ii) Primer: Two-component, epoxy mastic primer capable of creating a strong bond between concrete and adhesive mastic.

(iii) Adhesive Mastic: Two-component, high-solids, mastic adhesive and sealant. Adhesive shall be resistant to the environment in which the lining system is installed and shall create a strong bond between PVC sheet linings and primed concrete surfaces.

(iv) Seam Material: Plural-component, high-solids polyurethane for joining and sealing seams and terminations in PVC.

(v) Plastic Sheets and Shapes

1) For systems which use mechanically anchored sheets, PVC, as specified herein, with ribs on one side except for joint and repair strips which shall be plain on both sides; and

2) For systems which use chemically-bonded plastic sheets, PVC sheets and PVC pre-moulded corner pieces which shall be plain on both sides.

(vi) Activator: For chemically-bonded plastic sheet system, single component PVC primer especially formulated for bonding PVC sheets and PVC pre-molded corner pieces to the adhesive mastic and seam material.

(vii) Ultraviolet Blocker: Manufacturer’s recommended top coat for protection of PVC surfaces exposed to sunlight.

(c) PVC Products

(i) Resin: Manufactured from high molecular weight PVC resin and other necessary ingredients compounded to form a homogeneous, permanently flexible material for use as a protective lining in structures. PVC resin shall comprise not less than 99% by weight, and no recycled resin shall be used. Colour shall be white.

(ii) Forms: Sheets, strips, preformed interior and exterior corners, and other forms as required. Use maximum sheet size permitted by location and working conditions to minimize the number of seams. Preformed corners shall be nominally 75mm in each dimension.

(d) Water Stop and Void Repair Materials

(i) Use lining manufacturer's recommended materials to stop entrance of groundwater and to repair voids where required before beginning any lining work. Water stop and repair materials may be of cementitious or polymer based binder and aggregate, and shall be fully integrated with the lining system.

(e) Abrasive Materials

(i) Select abrasive type and size to produce a surface profile that meets manufacturer's recommendations for specific lining system to be applied.

G3.25.3 **Execution**

(a) General

(i) Inspection and Verification

1) Inspect and provide substrate surfaces prepared in accordance with these Specifications and the printed directions and recommendations of the manufacturer whose product is to be applied. The more stringent requirements shall apply.

(b) Protection of Items Not To Be Lined

(i) Remove, mask, or otherwise protect adjacent surfaces that will not receive lining. Protect items such as hardware, lighting fixtures, switch plates, aluminium surfaces, machined surfaces, couplings, shafts, bearings, nameplates on machinery, and other surfaces.

(ii) Protect working parts of mechanical and electrical equipment from damage during surface preparation and product application processes.

(c) Surface Preparation – Concrete A

(i) General

1) Do not begin until 30 days after concrete has been placed and has gained its design strength at 28 days or as required by the lining manufacturer.

2) Stop all leakage of groundwater or any other water source onto concrete surfaces before proceeding with any lining work.

3) Conform to Singapore Standard: Surface Preparation of Concrete, SSPC SP 13, unless otherwise required by the lining manufacturer.

4) Remove grease, oil, dirt, salts or other chemicals, loose materials, or other foreign matter by detergent or other suitable cleaning methods.

5) Blast clean to degree of cleanliness required by lining manufacturer. Follow manufacturer's recommendations for additional preparation if required for voids and air holes exposed after blasting.

6) Chemically treat concrete surfaces to achieve specific pH or other surface conditions required by lining manufacturer.

7) Measure the pH, temperature and moisture content of concrete surfaces to ensure that they are within lining manufacturer's recommended range.

8) Unless otherwise required for proper adhesion, ensure surfaces are dry prior to lining.

(d) Surface Preparation – Metal

(i) General: Prepare metal surfaces in concrete where they will be covered with the lining system. Round or chamfer sharp edges and grind smooth burrs, jagged edges, and prepare metal surfaces according to the following SSPC Specifications:

1) Solvent Cleaning: SP 1.

2) White Metal Blast Cleaning: SP 5.

(ii) Abrasive Blast Cleaning Requirements

1) Preblast Cleaning: Remove oil, grease, welding fluxes, and other surface contaminants by appropriate methods prior to blast cleaning.

2) Blast Cleaning: Attain specified degree of cleanliness. Minimum surface preparation is as specified herein and takes precedence over coating manufacturer's recommendations.

3) Select type and size of abrasive to produce a surface profile that meets lining manufacturer's recommendations.

(e) Installation of Anchored Sheet Lining System

(i) Priming: Apply primer to concrete surface in the manner and rate specified by the manufacturer and allow for this to cure as required prior to application of adhesive. Do not allow primed surfaces to stand beyond the window of time allowed by the manufacturer for application of the adhesive.

(ii) Adhesive Application: Trowel-apply adhesive mastic to the primed surface at the rate specified by the lining manufacturer. The surface area of adhesive application shall not exceed the area of work that can be lined within the pot life of the adhesive.

(iii) PVC Liner Application:

1) For mechanically anchored plastic sheet lining system, apply PVC lining while the bonding ability of the adhesive mastic is at its optimum:

- Align ribs on PVC sheets in a vertical direction to allow drainage of any moisture that may collect behind the lining;
- Align PVC sheets to permit installation of adhesive-bonded joints. Heat PVC as required to accommodate for corners and bends;
- After placing each sheet, use rollers and rubbing cloths as required to smooth and press the sheet in place so that it is lying flat against the concrete surface and uniformly bonded to the adhesive. Take care to ensure that all air pockets, wrinkles, and other surface regularities are removed from each sheet before the next sheet is applied;
- Install bonded joints using joint strips, seam sealant primer, and seam adhesive. Roll all joints to establish a tight bond between joint strips and PVC sheets; and

- Finish all terminations and closures, and seal all penetrations through the lining, as recommended by the manufacturer;
  - o For chemically-bonded plastic sheet lining system:
    - – Apply activator to PVC liner and pre-molded corners in accordance with manufacturer’s instructions. Activated PVC liner and pre-molded corner pieces shall be allowed to become tack-free before installation. Activated PVC sheets and pre-molded corners shall be protected from debris or contamination prior to installation.
    - – Apply activated PVC lining and pre-molded corners while the bonding ability of the adhesive mastic and seam material is at its optimum. Should bubbles greater than 150 mm in diameter, pinholes, or discontinuities form in the applied lining system, they shall be repaired per the manufacturer’s instructions.
    - – The liner seams shall be of the overlap type (See Detail X for Overlap Joint). Overlap seams shall be 100mm wide minimum, and shall be lapped in the direction of the flow, and lapped downwards for horizontal seams. All liner terminations and leading edges shall have a 6mm to 12 mm deep by 3mm to 6mm wide saw-cuts per manufacturer’s instructions. All liner seams, terminations, and leading edges shall be sealed using the seam material per manufacturer’s instructions. Resulting seam-overlay band shall be smooth, relatively flat, free of sharp protrusions, and have a thickness between 2mm and 5mm.
    - – Ultraviolet blocker shall be applied per manufacturer’s recommendations for protection of PVC surfaces exposed to sunlight.
  - o Trim and clean up any areas as required to produce a smooth, uniform lining with no loose edges or obvious defects. Remove excess adhesive at seams as recommended by the manufacturer. Visually inspect the installed lining to ensure completeness.

(iv) Curing

1) Provide cure time for lining as recommended by manufacturer for the specific site conditions. Do not immerse the lining prior to completion of the period.

2) Manufacturer’s representative shall inspect and certify that installed linings are cured and ready for service.

(v) Repairs

1) Repair pinholes, small cuts, and other point defects per manufacturer recommendations. All such repairs shall be finished by application of a patch of plain PVC sheet to overlap the adjacent lining by a minimum of 100mm.

2) Linings that fail the adhesion test and all air pockets larger than 150mm in diameter shall be cut out to remove the sheet and adhesive mastic. The exposed area shall be prepared according to the instruction of the lining manufacturer. Re-apply primer, adhesive, and sheets as required to properly repair the area. Large repairs shall overlap adjacent lining for a minimum of 100mm.

(f) Field Tests

(i) General

1) Conduct all tests in the presence of the S.O.. Provide facilities and assistance for S.O.'s final inspection including access, ventilation, staging and lighting.

2) Daily Field Reports: Provide completed field report forms provided by lining manufacturer to describe the types, conditions, and locations of lining work on a daily basis.

(ii) Adhesion Testing

1) For mechanically anchored plastic sheet lining system:

- Perform adhesion tests as specified herein and as required by the S.O. to ensure that the bond strength of the lining conforms to all requirements. Perform a minimum of one adhesion test per 100m2 of lined surface. Perform additional adhesion tests at seams at the minimum rate of one test per 200m of seam; and
- Conform to ASTM C794 as modified for field test conditions. Prepare 25mm wide pulling strips during the lining operation and conduct tests as recommended by the lining manufacturer. Test results shall demonstrate that installed lining meets or exceeds manufacturer's ratings in 100% of the tests after the recommended curing time for the site conditions.

2) For chemically-bonded plastic sheet lining system:

- Perform ASTM D4541 pull-off adhesion tests per the manufacturer's recommendations and as required by the S.O. to ensure that the bond strength of the lining conforms to all requirements. Perform a minimum of one adhesion test per 100 square meters of lined surface. The area to be tested shall be cored through the lining system past the bond area with the concrete substrate. The allowable minimum value for the pull-off strength test shall be per the protective lining system manufacturer's recommendation, but shall not be less than 2.5 Bar after three (3) days of curing at a minimum ambient temperature of 25°C.

3) Spark (Pinhole) Testing (For mechanically anchored plastic sheet lining system only):

- Conduct spark testing of 100% of the lined surfaces with high voltage pinhole detectors recommended by the lining manufacturer and in accordance with NACE RP0188. Test voltage shall be 20,000 volts minimum or higher as recommended by manufacturer; and

- All defects shall be marked for repair using a permanent marker.

(iii) Repetition of Tests: Repair all defects identified by the testing and re-test all repaired areas until 100 % of the lining meets all requirements.

(g) Ultraviolet Protection

(i) Provide coating for protection against ultraviolet light on all lining surfaces that are exposed to sunlight in normal service.

END OF SECTION